

WORKFORCE MANAGEMENT SYSTEM: ATTENDANCE, SHIFT, AND LEAVE TRACKING

Faculty

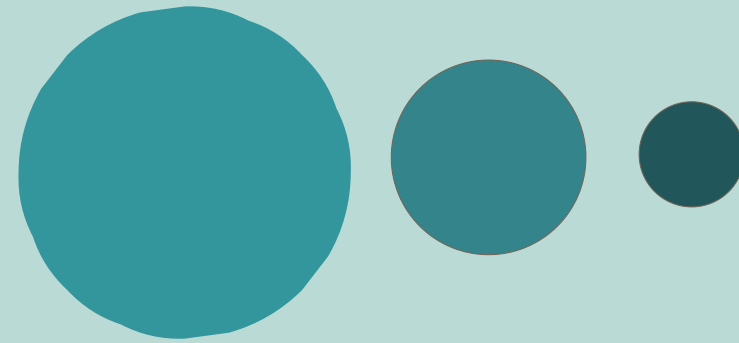
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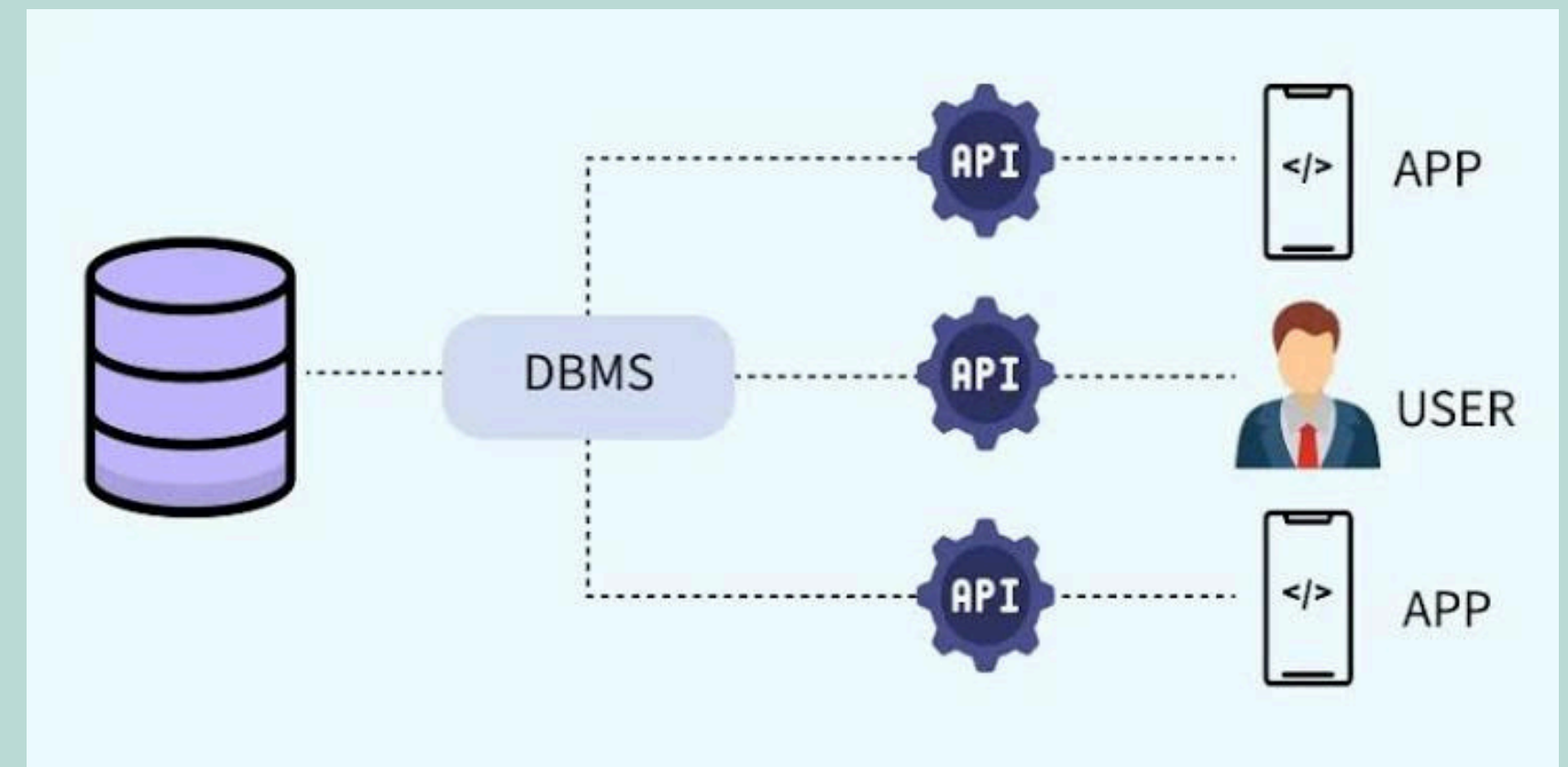
Overview



Design and Implementation of a Database Management System

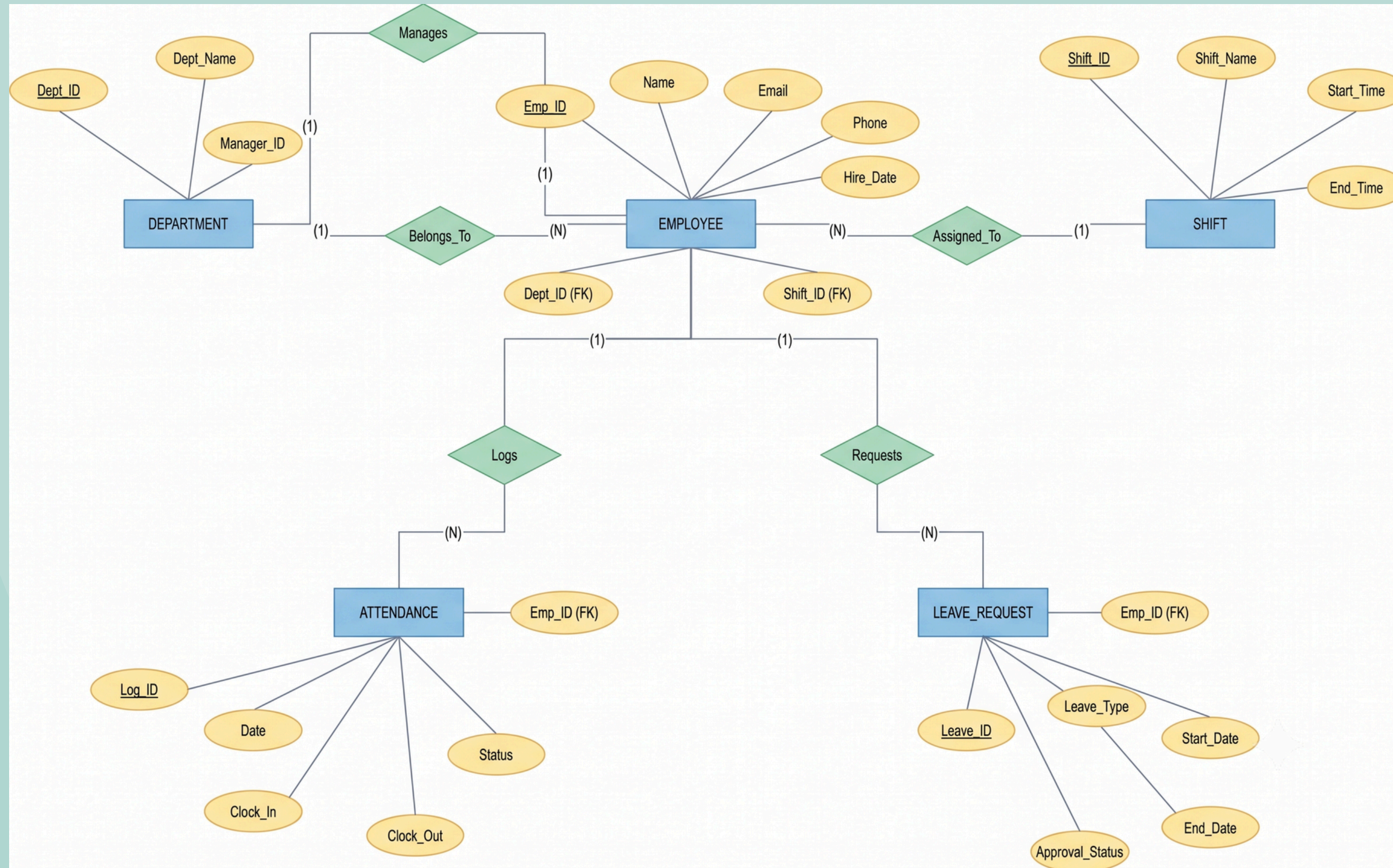
The Workforce Management System is a centralized relational database designed to automate and streamline core HR operations, specifically focusing on employee attendance, shift scheduling, and leave management. By replacing fragmented manual tracking methods with a unified architecture, the system ensures data integrity and enforces critical business rules, such as validating daily work hours and managing leave approval statuses.

Ultimately, this robust database provides a single, reliable source of truth that simplifies administrative workflows, optimizes workforce coverage, and enhances overall organizational efficiency.

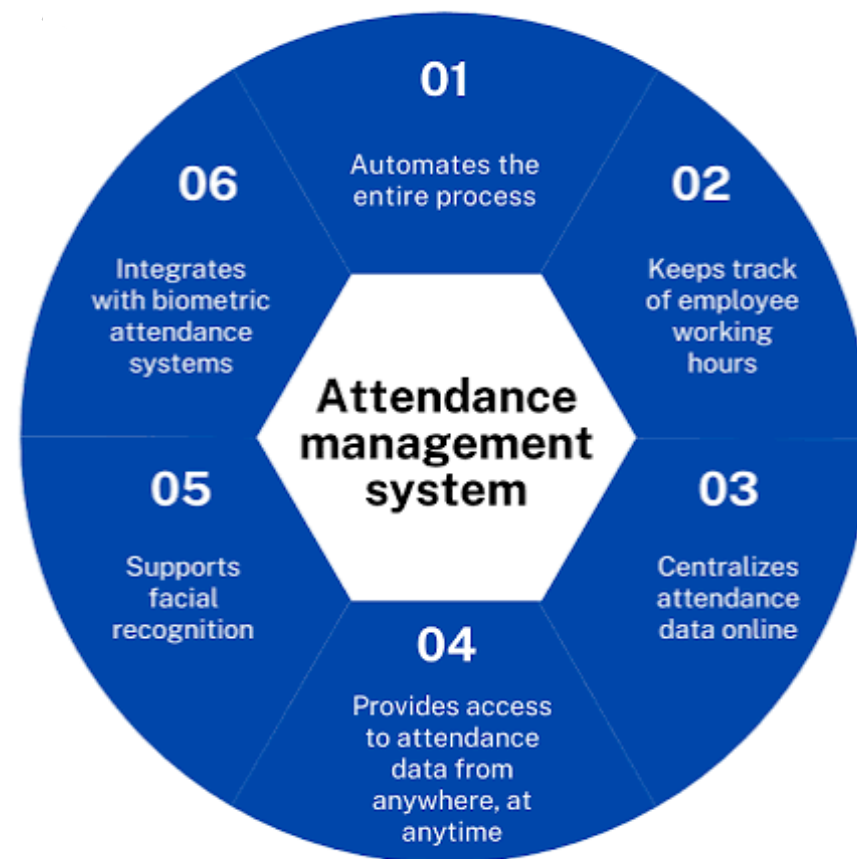




ER DIAGRAM



ENTITIES FOR THE WORKFORCE MANAGEMENT SYSTEM



- DEPARTMENT
- SHIFT
- EMPLOYEE
- ATTENDANCE
- LEAVE_REQUEST

DATABASE SCHEMA: CORE ENTITIES AND ATTRIBUTES

Employee

Emp_ID
Name
Email
Phone
Hire_Date
Dept_ID
Shift_ID

Department

Dept_ID
Dept_Name
Manager_ID

SHIFT

Shift_ID
Shift_Name
Start_Time
End_Time

ATTENDANCE

Log_ID
Emp_ID
Date
Clock_In
Clock_Out
Status

LEAVE

Leave_ID
Emp_ID
Leave_Type
Start_Date
End_Date
Approval_Status

DATABASE SCHEMA: CORE ENTITIES AND ATTRIBUTES

Employee

- Emp_ID (INT, **Primary Key**, Auto-Increment, Not NULL)
- Name (VARCHAR, Not NULL)
- Email (VARCHAR), Unique, Not NULL)
- Phone (VARCHAR, Unique, Not NULL)
- Hire_Date (DATE, Not NULL)
- Dept_ID (INT, Foreign Key referencing DEPARTMENT.Dept_ID, Not NULL)
- Shift_ID (INT, **Foreign Key** referencing SHIFT.Shift_ID, Not NULL)

Department

- Dept_ID (INT, **Primary Key**, Auto-Increment, Not NULL)
- Dept_Name (VARCHAR, Unique, Not NULL)
- Manager_ID (INT, Not NULL)

SHIFT

- Shift_ID (INT, **Primary Key**, Auto-Increment, Not NULL)
- Shift_Name (VARCHAR, Unique, Not NULL)
- Start_Time (TIME, Not NULL)
- End_Time (TIME, Not NULL)

DATABASE SCHEMA: CORE ENTITIES AND ATTRIBUTES

ATTENDANCE

- Log_ID (INT, **Primary Key**, Auto-Increment, Not NULL)
- Emp_ID (INT, **Foreign Key** referencing EMPLOYEE.Emp_ID, Not NULL)
- Date (DATE, Not NULL)
- Clock_In (TIME, Not NULL)
- Clock_Out (TIME, NULL allowed – for active shifts)
- Status (VARCHAR, Not NULL)

LEAVE

- Leave_ID (INT, **Primary Key**, Auto-Increment, Not NULL)
- Emp_ID (INT, **Foreign Key** referencing EMPLOYEE.Emp_ID, Not NULL)
- Leave_Type (VARCHAR, Not NULL)
- Start_Date (DATE, Not NULL)
- End_Date (DATE, Not NULL)
- Approval_Status (VARCHAR, Not NULL)



THANK
YOU

